

TAIVALKOSKI KK URHEILUTIE, Finland

WMO: 028040

Lat: 65.575N

Lon: 28.218E

Elev: 646

StdP: 14.36

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.5	-21.1	-32.5	0.9	-26.1	-26.7	1.3	-21.0	17.2	27.1	15.7	26.1	1.5	50	0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	78.8	62.2	74.6	60.6	71.1	59.5	65.7	73.8	63.5	71.5	61.3	68.4	7.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	88.3	68.0	60.3	80.1	66.3	58.1	73.7	64.4	30.9	74.3	29.1	71.2	27.6	68.8	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	14.2	12.5	DB	-30.1	80.9	6.4	4.7	-34.8	84.2	-38.5	87.0	-42.2	89.6	-46.8	93.0	(4)
(5)			WB	-29.6	67.2	5.9	3.2	-33.8	69.5	-37.3	71.4	-40.6	73.2	-44.8	75.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.2	9.6	13.2	19.4	31.5	44.6	52.7	59.3	54.7	46.4	34.1	25.6	17.4		
	DBStd	19.28	14.28	14.40	10.98	8.14	8.58	7.41	6.48	6.07	6.21	7.46	9.72	13.63		
	HDD50	6467	1253	1031	949	557	212	54	7	24	140	495	733	1012		
	HDD65	11292	1718	1451	1414	1006	633	374	197	322	558	959	1183	1477		
	CDD50	681	0	0	0	0	45	136	296	171	32	1	0	0		
	CDD65	31	0	0	0	0	0	7	21	3	0	0	0	0		
	CDH74	400	0	0	0	0	26	89	240	45	0	0	0	0		
	CDH80	59	0	0	0	0	0	14	40	5	0	0	0	0		
	WSAvg	6.0	5.2	5.8	6.3	6.3	6.8	6.5	5.6	5.6	5.9	6.1	6.0	5.8		
	PrecAvg	27.1	1.9	1.6	1.5	1.3	2.2	2.8	3.7	2.9	2.4	2.4	2.3	2.2		
(15) Precipitation	PrecMax	36.9	3.3	3.0	2.6	2.3	4.1	5.2	5.6	6.1	4.4	5.0	4.2	3.6		
	PrecMin	21.7	0.6	0.4	0.6	0.4	0.5	1.1	1.6	1.2	0.3	0.5	0.6	0.7		
	PrecStd	3.9	0.7	0.7	0.6	0.6	0.7	1.1	1.2	1.2	1.1	1.0	0.9	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	38.7	42.0	59.9	77.3	82.0	84.3	79.8	66.6	51.8	44.0	35.7		
		MCWB	34.0	36.1	36.2	46.8	56.3	62.3	66.1	65.7	58.0	50.3	42.9	35.0		
	2%	DB	32.2	35.2	38.6	51.0	73.0	76.1	80.2	73.5	61.9	47.9	40.9	34.3		
		MCWB	31.7	33.7	33.9	42.7	55.9	60.8	63.9	62.3	54.8	45.9	39.8	33.5		
	5%	DB	30.4	33.1	35.6	46.9	67.0	70.4	76.2	70.0	59.2	45.8	37.9	33.3		
		MCWB	30.0	32.2	32.5	38.9	53.6	57.6	62.3	60.2	53.3	44.2	37.2	32.6		
	10%	DB	27.6	30.9	33.7	43.5	60.8	66.6	72.4	66.1	56.4	43.7	35.6	32.1		
		MCWB	27.0	30.2	31.6	37.1	49.7	55.6	61.0	58.3	52.4	42.3	35.1	31.6		
	0.4%	WB	34.2	36.2	37.5	47.9	59.6	65.7	69.1	67.4	59.4	50.7	43.4	35.1		
		MCDB	34.9	37.8	41.0	56.6	71.1	73.8	75.9	73.9	64.6	51.7	44.1	35.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.9	34.0	34.8	42.8	56.7	63.0	66.5	64.0	56.3	46.8	40.0	33.8		
		MCDB	32.1	35.2	37.0	49.6	70.2	73.4	74.5	70.6	59.8	47.5	40.7	34.1		
	5%	WB	30.0	32.2	33.3	39.7	53.9	60.1	64.7	61.5	54.5	44.4	37.3	32.7		
		MCDB	30.4	32.9	34.9	44.6	66.6	68.3	73.1	67.1	58.0	45.4	37.9	33.1		
	10%	WB	27.1	30.2	31.8	38.0	50.5	56.9	62.6	59.4	52.8	42.3	35.1	31.6		
		MCDB	27.6	30.9	33.4	42.2	58.9	64.0	70.3	64.5	55.8	43.7	35.4	32.1		
	5% DB	MDBR	13.2	14.8	19.1	17.6	18.6	17.1	17.6	16.6	13.3	9.2	8.6	11.1		
		MCDBR	10.2	10.3	15.4	23.0	30.0	24.9	25.2	23.6	19.0	10.9	7.8	7.5		
		MCWBR	10.4	9.8	12.4	14.4	15.0	11.2	11.2	12.8	12.3	9.3	7.8	7.4		
		MCWBR	10.2	9.9	10.9	18.8	27.4	20.2	20.6	19.2	14.8	10.3	7.5	7.2		
(38) Clear-Sky Solar Irradiance	taub		0.200	0.269	0.300	0.334	0.343	0.343	0.362	0.342	0.318	0.292	0.351	0.201		
		taud	1.930	2.102	2.150	2.231	2.449	2.475	2.440	2.477	2.528	2.365	2.648	1.958		
	Ebn at Noon		84	188	240	262	274	276	266	261	243	187	83	16		
		Edh at Noon	6	18	28	33	30	30	30	26	20	14	6	2		
(44) All-Sky Solar Radiation	RadAvg		32	188	642	1195	1492	1621	1524	1114	603	208	47	7		
	RadStd		3	37	59	116	142	125	151	110	62	32	8	1		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)		+1.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page